

CE – 101 : C & P F*Computing and Programming
Fundamentals***Batch 2015**

Sir Syed University of Engineering & Technology
Computer Engineering Department
University Road, Karachi-75300, PAKISTAN

*Compiled By:**Syed Atir Iftikhar***CE - 101 : Computing and Programming
Fundamentals****■ Course Objectives:**

- Upon successful completion of this course, the student will be able to:
 - Fundamentals of Computer Engineering and Information Technology
 - Overview of History, Classification and Components of Computers
 - Number Systems and Logic Gates
 - Overview of Software, Operating Systems, Networks and Internet
 - Introduction about Programming
 - Basic Building Blocks, Loop, Decision Making

C & P F

C&PF Books

▪ Text Book:

1. Introduction to Computers (7th Edition)

By Peter Norton

2. Turbo C Programming For The PC (Revised Edition)

By Robert Lafore

▪ Reference Books:

1. Computer, Communications and Information.

By Sarah Hutchinson and Stacey Sawyer

2. Let Us C

By Yashavant Kanetkar

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Marks Distribution

▪ Mid Term _____ 20

▪ Lab Work _____ 20

▪ Quiz _____ 5

▪ Assignment + Class Performance _____ 5

▪ Semester Final Paper _____ 50

▪ Total Marks _____ 100

▪ *Performance Bonus* _____ *5 Marks*

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C & P F**Course Instructors**▪ **Syed Aatir Iftikhar**

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Course Outline**✓ PART A**

- **History of Computer**
- **Classification of Computer**
- **Basic Components**
- **CPU, I/O, Peripheral Devices, Storage**
- **Von Neumann Architecture**
- **Number Systems**
- **Binary Numbers**
- **Boolean Logic**
- **Software**
- **Operating Systems**
- **Networks and Internet**

✓ PART B

- **Types of Programming Languages**
- **Algorithm**
- **Flow Chart**
- **Introduction to C Language Programming**
- **Basic Building Blocks**
- **Loops**
- **Decision Making Statements**

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CE – 101: Computing and Programming Fundamentals

Chapter**1****Batch 2014****Introduction To Computers****Learning Objectives**

1. What is a Computer
2. Input – Processing – Output
3. Parts of Computer
4. History of Computer
5. Generations of Computer
6. Types of Parts of Computer

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What is a Computer ?

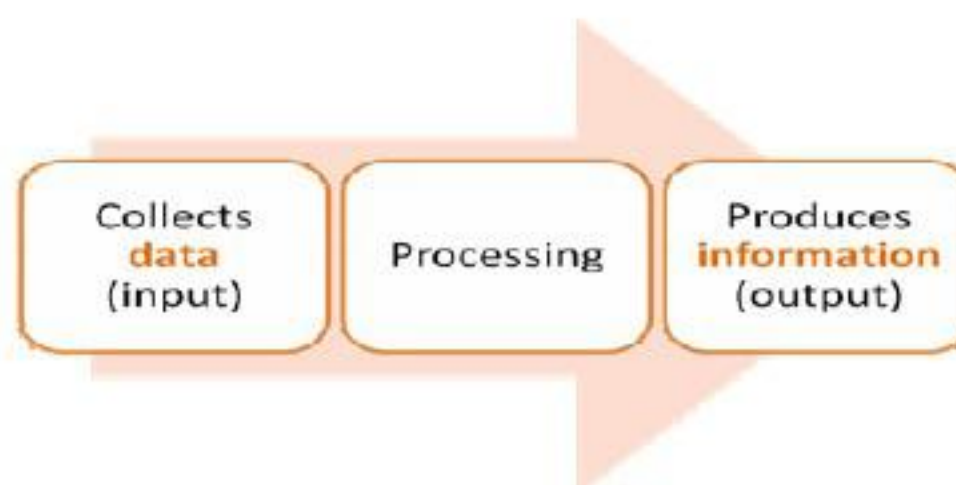
- A data processing machine operated automatically under the control of a list of the instructions (called a program) stored in its main memory.
- A Computer is a smart or intelligent machine, which can work only on certain instructions given by a human being.

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- A **computer** is an electronic device, operating under the control of instructions stored in its own memory

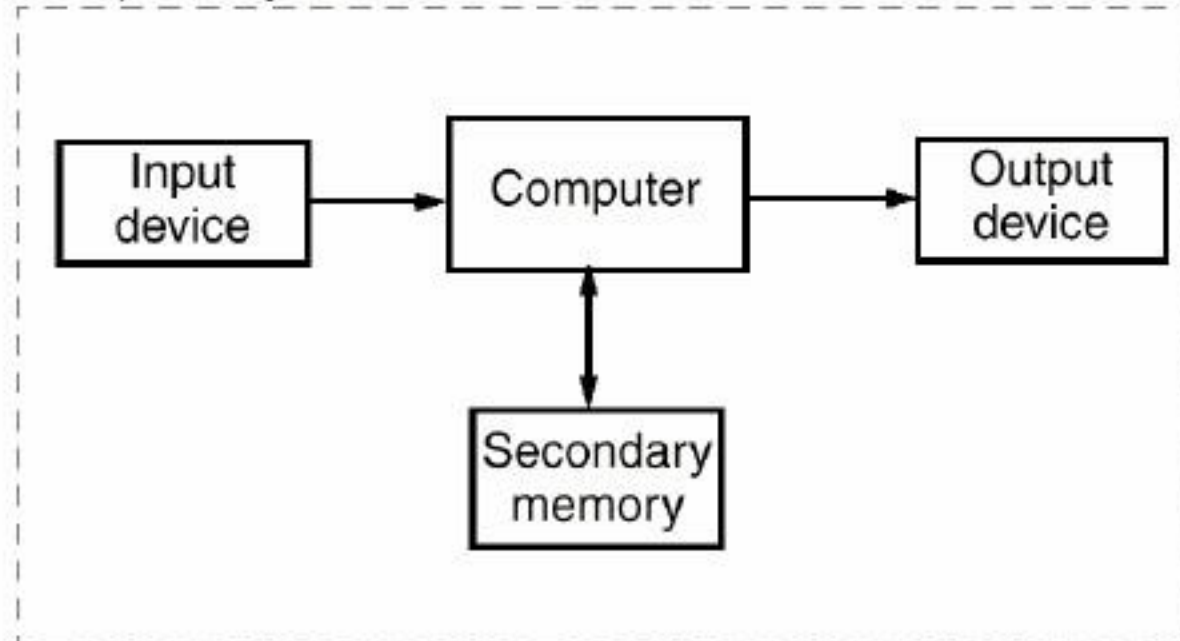
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What is a Computer ?

Computer system



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Components of a Computer

Input Device

- Allows you to enter data and instructions into a computer

Output Device

- Hardware component that conveys information to one or more people

System Unit

- Case that contains the electronic components of the computer that are used to process data

Storage Device

- Holds data, instructions, and information for future use

Communications Device

- Enables a computer to send and receive data, instructions, and information to and from one or more computers or mobile devices

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Input - Processing - Output▪ **Input**

What you type, read, or enter into computer

Devices

- Keyboard, Mouse, Scanner, Sensors, etc

▪ **Processing**

Changes the input data via formatting, sorting, and calculations

Devices

- Components on the motherboard, CPU, ALU, Memory

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Input - Processing - Output▪ **Output**

Results of computer processing

Devices

- Screen, Printer, Plotter, Signals, etc.

▪ **Storage**

Save information for later processing

Devices

- Memory, Hard Disk Tape Disk, USB, etc.

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Input - Processing - Output



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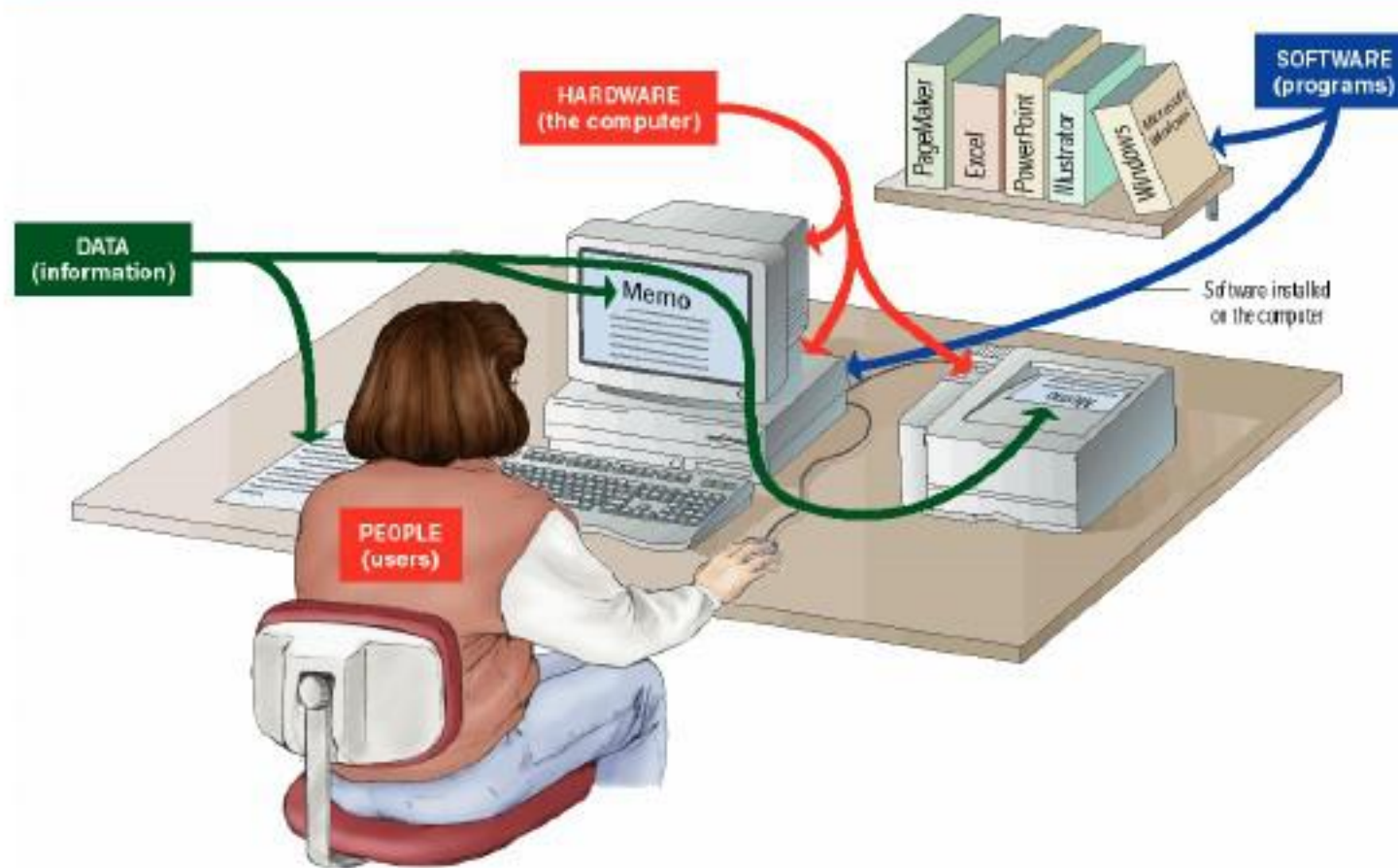
Parts of a Computer System

- **What is a Computer?**
 - A computer is an electronic device used to process data.
 - A computer can convert data into information that is useful to people.
 - A complete computer system includes 4 distinct parts:
 1. Hardware
 2. Software
 3. Data
 4. Users

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Parts of a Computer System



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Parts of a Computer System

Hardware

- A computer's hardware consists of electronic devices; the parts you can see and touch.
- The term "device" refers to any piece of hardware used by the computer, such as a keyboard, monitor, modem, mouse, etc.

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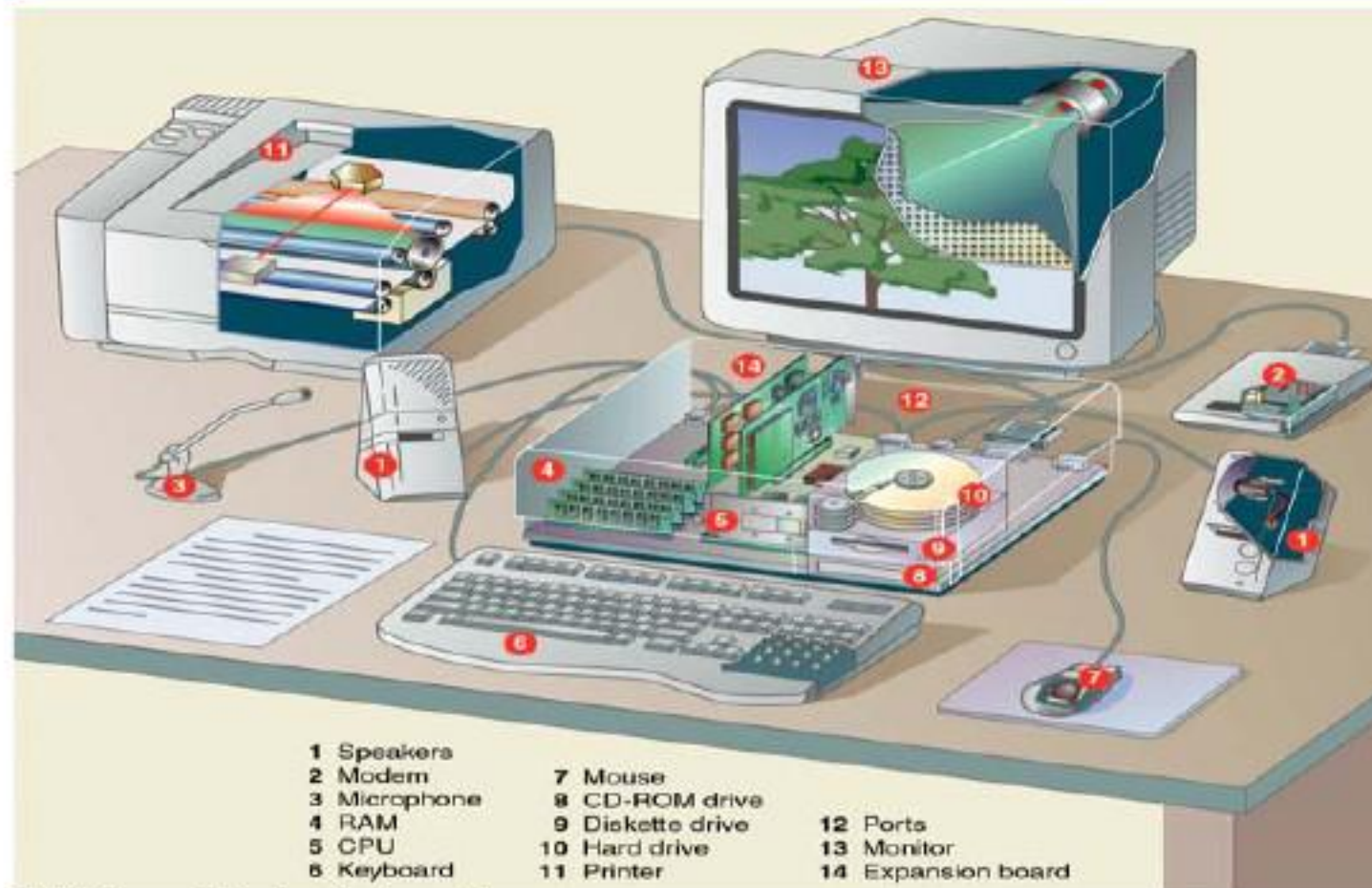
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Parts of a Computer System



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Parts of a Computer System

■ Software

- Software – also called Programs – consists of organized sets of instructions for controlling the computer.
- Some programs exist for the computer's use, to help it manage its own tasks and devices.
- Other programs exist for the user, and enable the computer to perform tasks for you, such as creating documents.

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Parts of a Computer System

▪ Data

- Data consists of raw facts, which the computer can manipulate and process into information that is useful to people.
- Computerized data is digital, meaning that it has been reduced to digits, or numbers. The computer stores and reads all data as numbers.
- Although computers use data in digital form, they convert data into forms that people can understand, such as text, numerals, sounds, and images.

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Parts of a Computer System

▪ People

- People are the computer's operators, or users.
- Some types of computers can operate without much intervention from people, but personal computers are designed specifically for use by people.

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History of Computer

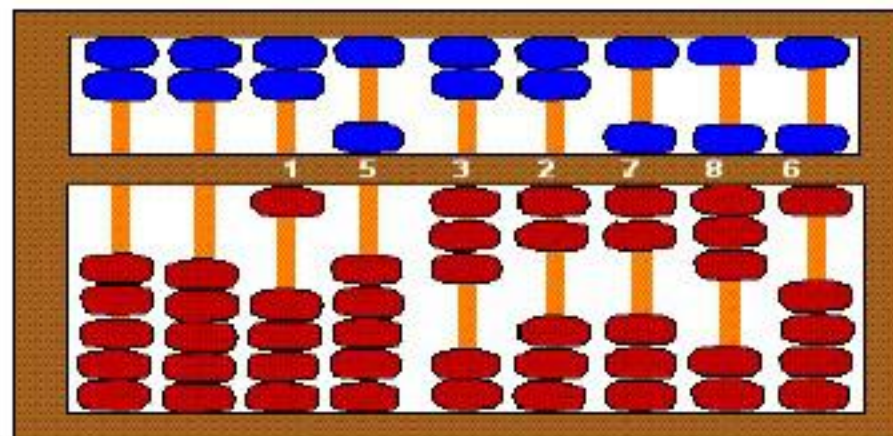
- **Abacus** (16th century)
An early device to record numeric values
- **Blaise Pascal** (middle 17th century)
Mechanical (gear driven) device to add, subtract, divide & multiply
- **Joseph Jacquard** (late 18th century)
Jacquard's Loom, the punched card
- **Charles Babbage** (19th century)
Analytical Engine, designed but never implemented
- **Alan Turing**
Turing Machine, Artificial Intelligence Testing

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History of Computer

- **Abacus** (16th century)
- The **abacus** also called a **counting frame**, is a calculating tool used primarily for performing arithmetic processes.



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History of Computer

- **Blaise Pascal** (middle 17th century)
- Blaise Pascal invented the mechanical calculator in 1642 which perform add and subtract two numbers directly and multiply and divide by repetition.



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History of Computer

- **Joseph Jacquard** (late 18th century)
- The Jacquard Loom is a mechanical loom that has holes punched in pasteboard, each row of which corresponds to one row of the design.



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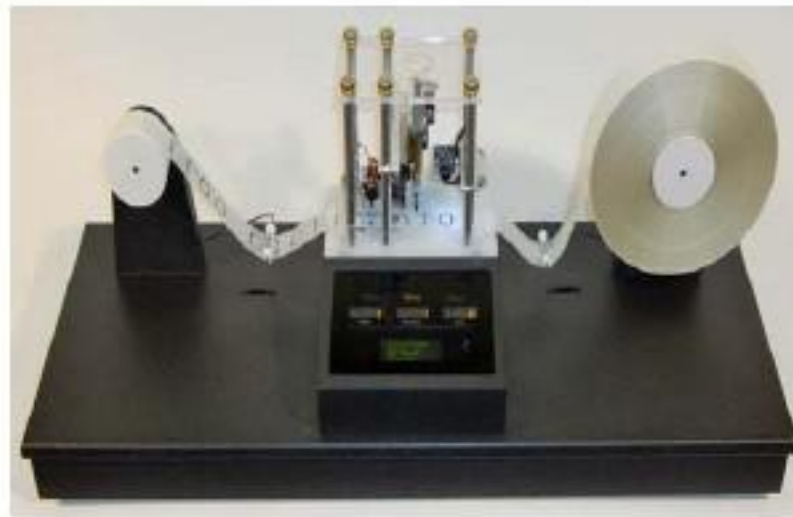
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History of Computer

- **Alan Turing** (19th century)
- Turing describes a machine that has an infinitely long tape upon which it writes, reads and alters symbols. The characters, ones and zeros, are written by the machine with a black dry erase marker.



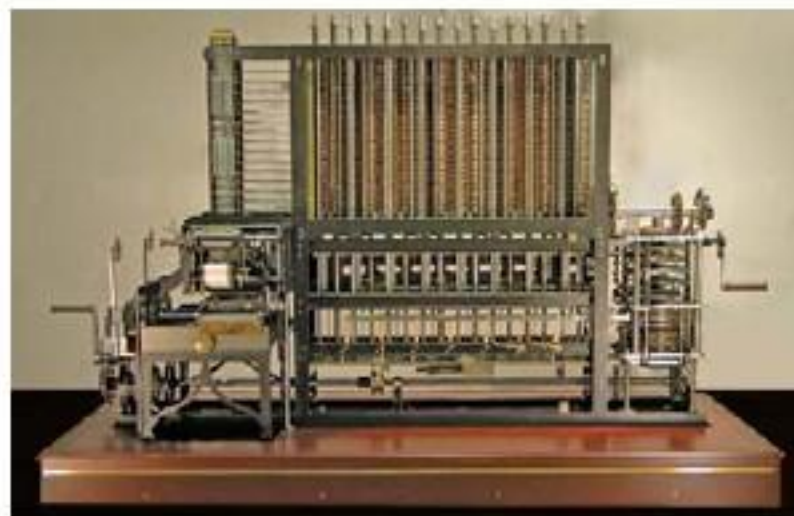
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History of Computer

- **Charles Babbage** (19th century)
- **Charles Babbage**, considered a "father of the computer", Babbage is credited with inventing the first mechanical computer that eventually led to more complex designs.



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Generations of Computer

- Vacuum Tube 1946-1957
- Transistor 1958-1964
- Small Scale Integration 1965-1968
 - up to 100 devices
- Medium Scale Integration 1968-1971
 - 100-3,000 devices on a chip
- Large Scale Integration - 1972-1977
 - 3,000 - 100,000 devices on a chip
- Very Large Scale Integration - 1978 to date
 - 100,000 - 100,000,000 devices on a chip
- Ultra Large Scale Integration
 - Over 100,000,000 devices on a chip

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Generations of Computer

Generation	Approximate Dates	Technology	Typical Speed (Operation / Second)
1	1946-1957	Vacuum Tube	40,000
2	1958-1964	Transistor	200,000
3	1965-1971	Small and Medium Scale Integration	1,000,000
4	1972-1977	Large Scale Integration	10,000,000
5	1978-Onwards	Very Large Scale Integration	100,000,000

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Generations of Computer

▪ Vacuum Tube

- A **vacuum tube**, **electron tube** is a device controlling electric current through a vacuum in a sealed container. The container is often thin transparent glass in a roughly cylindrical shape.
- They are really big, get very hot and use a lot of electricity.



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Generations of Computer

▪ Transistor

- Transistor circuits are designed based on current flow, while tube circuits are designed based on voltages.
- Transistors are solid and based on some mixture of silicon.
- Transistor are very small, generate almost no heat and use very little electricity.



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Generations of Computer

- SSI, MSI, LSI, VLSI
- An **Integrated Circuit** (also referred to as an IC, a **chip**, or a **microchip**) is a set of electronic circuits on one small plate ("chip") of semiconductor material, normally silicon.



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Generations of Computer

- **Small-Scale Integration (SSI), Medium-Scale Integration (MSI), Large-Scale Integration (LSI), Very-Large-Scale Integration (VLSI)**
- Integrated circuits are used in virtually all electronic equipment today and have revolutionized the world of electronics. Computers, mobile phones, and other digital home appliances are now inextricable parts of the structure of modern societies, made possible by the low cost of producing integrated circuits.

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Generations of Computer

- The first integrated circuits contained only a few transistors. Called "**Small-Scale Integration**" (SSI), digital circuits containing transistors numbering in the tens provided a few logic gates
- The next step in the development of integrated circuits, taken in the late 1960s, introduced devices which contained hundreds of transistors on each chip, called "**Medium-Scale Integration**" (MSI).
- Further development, driven by the same economic factors, led to "**Large-Scale Integration**" (LSI) in the mid 1970s, with tens of thousands of transistors per chip.

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Generations of Computer

- **Very-large-scale integration (VLSI)** is the process of creating integrated circuits by combining thousands of transistors into a single chip. VLSI began in the 1970s when complex semiconductor and communication technologies were being developed. The microprocessor is a VLSI device.



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Generations of Computer

Scale of Integration	Number of Transistors
SSI	$2 \approx 64$
MSI	$64 \approx 2000$
LSI	2000 - 64,000
VLSI	64,000-2,000,000
ULSI	2,000,000 - No limit

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Types of Computers

- Computers are of four types:
 1. Micro Computers
 2. Mini Computers
 3. Main Frames
 4. Super Computers

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Types of Computers

1. Micro Computers

- This most widely used computer generally employs a microprocessor, “computer on a chip” and are desktop sized or less
- Two main types
 - Desktop (PC's & Workstations)
 - Portable (Notebooks, Palmtops)

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Types of Computers

1. Micro Computers

- **Desktop**
(PC's & Workstations)
 - Small enough for a desktop, but not easily portable
 - Personal computers (PC's) run general purpose software and are employed by a wide spectrum of users



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Types of Computers

1. Micro Computers

- **Portable**
(Notebooks, Palmtops)
 - Easily transported from one place to another
- Four categories
 1. Laptops
 2. Notebooks
 3. Palmtops
 4. Personal Digital Assistants (PDA)



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Types of Computers

2. Mini Computers

- Desk-sized
- More processing speed and storage capacity than microcomputers
- General data processing needs at small companies
- Larger companies use them for specific purposes



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Types of Computers

3. Main Frames

- Larger machines with special wiring and environmental controls
- Faster processing and greater storage than minicomputers
- Typical machine in large organizations

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Types of Computers

4. Super Computers

- The most powerful of the four categories
- Used by very large organizations, particularly for very math-intensive types of tasks

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